



Optimization Techniques for LLM Inference on a Multi-NPU System

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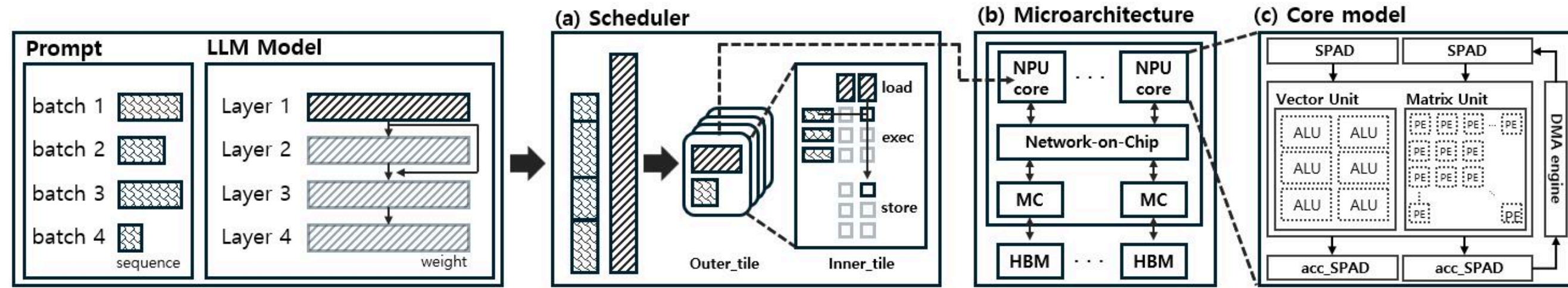
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Objective

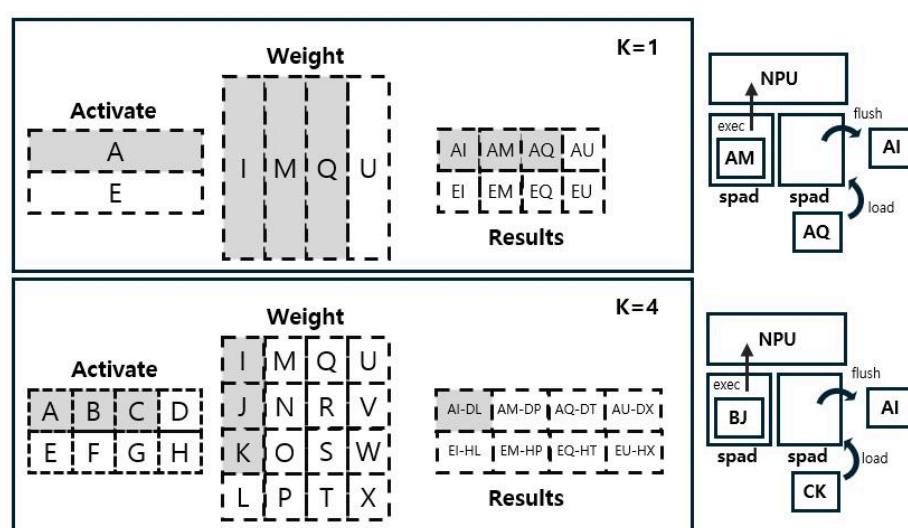
This research aims to (1) increase Core Utilization on LLM Inference by analyzing the major bottlenecks occurring in multi-NPU System and (2) propose novel NPU execution optimization techniques that holistically improve data movement.



Introduction

(a) scheduler

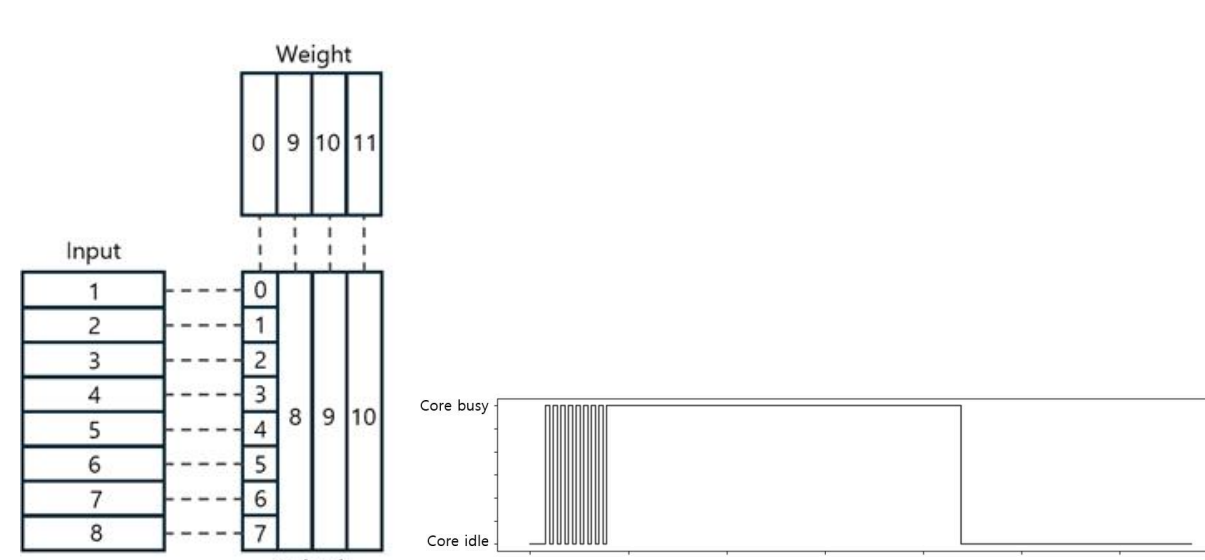
(1) Redundant activation fetches across tiles and cores



Repeated **loading and flushing of the same activation** causes significant inefficiency, and when the layer has many weights ($K \geq 4$), **SPAD resource constraints** make **activation flushing unavoidable**. Moreover, all cores requesting the same activation simultaneously leads to additional overhead.

(a) scheduler

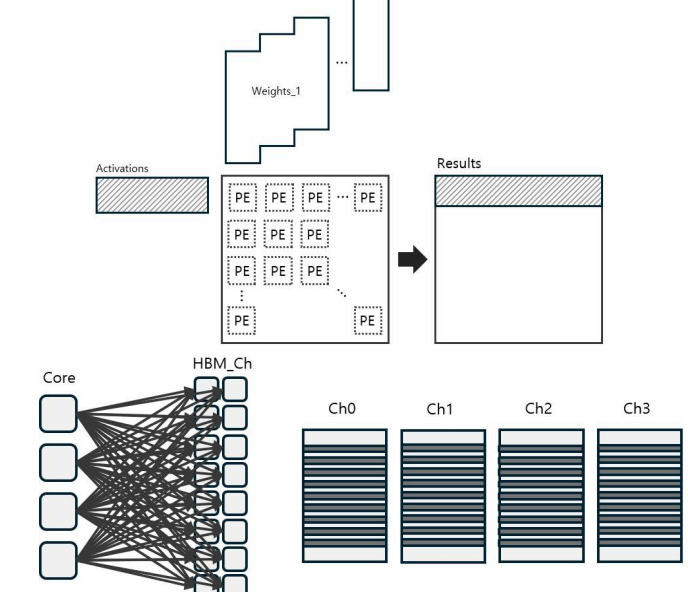
(2) Core Idling from Scheduling and Imbalanced Tile Distribution



Tile generation involves many factors that must be considered simultaneously, **making it challenging** to design an algorithm that **assigns balanced work** across cores. Furthermore, the load-execution scheduling is **not optimized to hide memory latency**.

(b,c) Microarchitecture, Core model

(3) Idle PE unit in Decoding and row buffer contention in multi-Core system

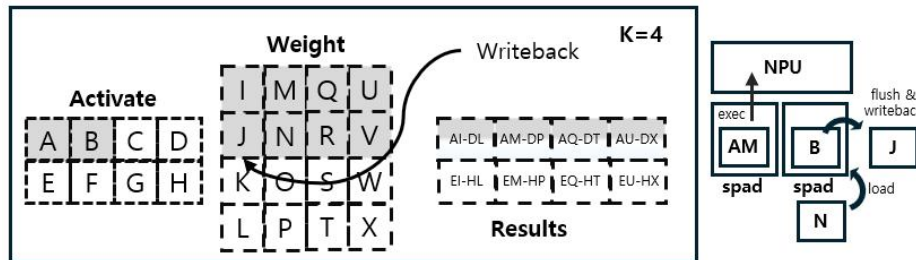


In decoding, all PEs are active, but **only a small portion produce meaningful results**. The high-performance core is significantly underutilized. In **multi-NPU environments**, HBM suffers from **severe row-buffer contention**, limiting effective bandwidth.

Experiments

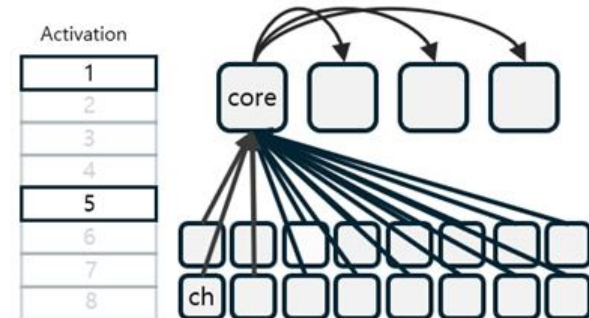
(1) Reduction of repeated data access

• Early writeback



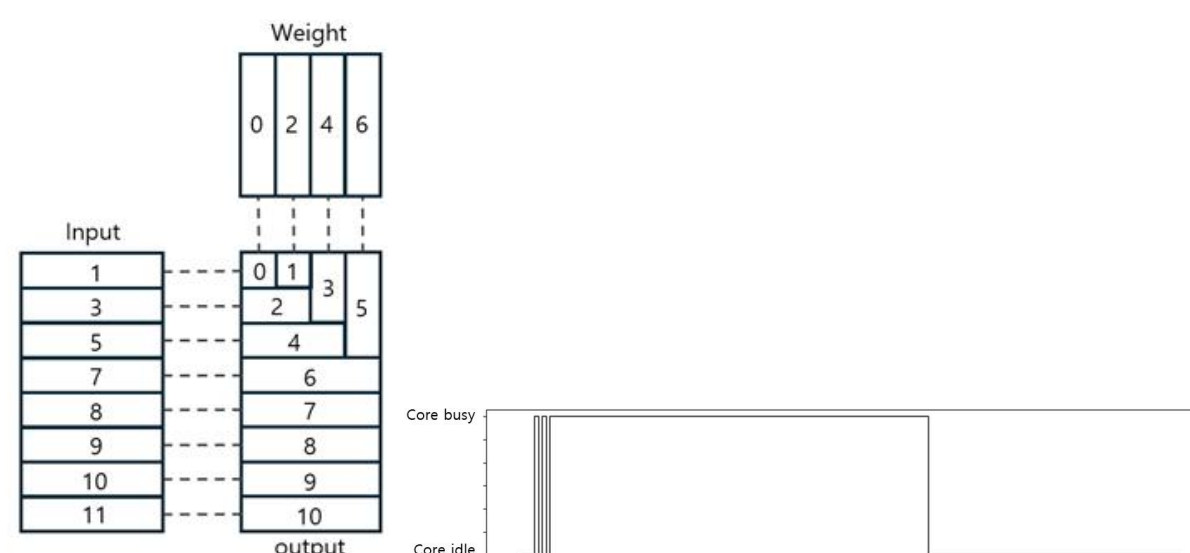
The core can reuse activations. Although **writebacks double**, a **double buffer can hide the write latency**.

• Sharing with Inter-Core Network



Cores request the same activations. To avoid redundant requests, **each core fetches a portion of the activations and shares it via the Inter-Core network**.

(2) Load Balancing and Hiding memory latency



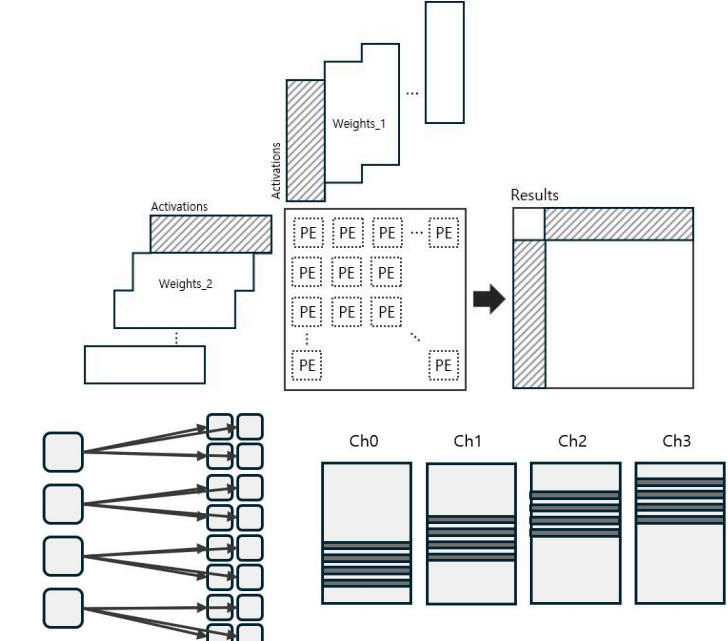
• Outer-Tile

For optimal utilization of Acc-SPAD and SPAD, **relax the rigid 1:1 activation-weight ratio**. To achieve load balancing, **ignore K-axis tiles** in the tile-distribution analysis.

• Inner-Tile

A scheduling policy that **reorders load and execution** can effectively **hide memory latency** and reduce core idle time.

(3) Mitigation of idle compute units and Alleviation of row buffer contention

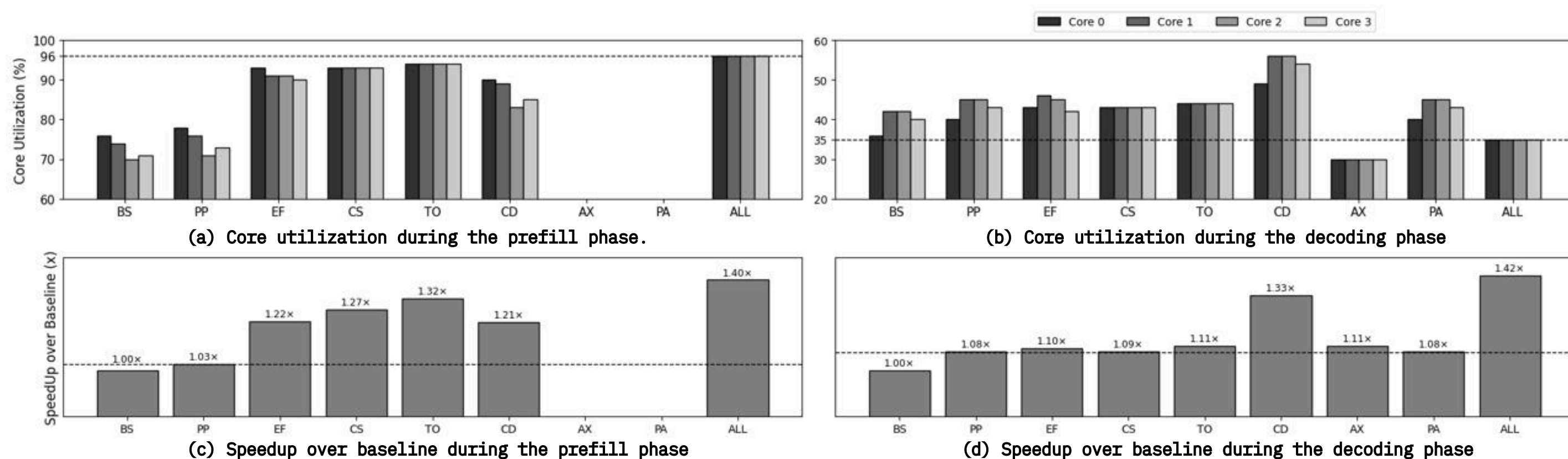


By placing weights in previously unused regions, **both axes of the systolic array can be fully utilized**. This **doubles the number of PEs producing meaningful results** and shortens execution time. and **by isolating HBM channels** as in a single-core environment, **row-buffer contention is dramatically reduced**, enabling near-maximum HBM bandwidth utilization.

Evaluation

Table 1: Configuration of ONNXim Platforms

NPU configuration		HBM Organization	
Model	TPU	Model	HBM2
Systolic size/Chip	128 x 128	Channel	16
Vector Unit/Chip	65536-bit	Bank/BankGroup	4
SPAD size	32 MB	Bank/Channel	32
Acc SPAD size	4 MB	Capacity/Channel	1GB
Frequency	1GHz	Frequency	1.2GHz
HBM Timing Parameter (Cycle)		Network Configuration	
nBL=2, nCL=9, nRCRD=9, nRCDDR=9, nRP=9, nCWL=3, nWTRS=4, nWTRL=5, nRTPL=4		Topology	Fly
		Frequency	1GHz



BS(Baseline), PP(HBM Pre-charge Policy), EF(Eliminate Unnecessary Spad Flush), CS(Inter-Core Network Sharing), TO(Tiling Optimization), CD(Channel Distribution), AX(2-Axis Execution), PA(Parallel Attention), ALL(All Optimizations Applied)

• Prefill

Core Utilization : 75%→96%

Speedup : 1.40x

• Decoding

Core Utilization : 43%→35%

Speedup : 1.42x

Conclusions

This study observed that the key bottleneck determining performance in Multi-NPU based LLM Inference Services is not the computational capacity of the NPU but rather data movement. We proposed an integrated optimization methodology for data reuse, alleviating memory access, integrated tiling policy, and mitigated row buffer contention and utilization idle PE unit within a multi-NPU System. By achieving maximal enhancement of Speedup and Core Utilization through optimization techniques, this research suggests an effective strategy for practical performance improvement in NPU-based LLM inference. Overall, by addressing issues such as unnecessary data movement, inefficient data scheduling, and underutilization of PE units, this study achieved a speedup of 40% in the Prefill phase and 43% in the Decoding phase on the Llama3-8B model using a 4-core NPU system. Furthermore, core utilization in the Prefill phase increased from 76% to 96%, while the Decoding phase achieved performance gains without a commensurate increase in core utilization, demonstrating improved execution efficiency. These results indicate that enhancing the end-to-end dataflow, from memory, to SPAD, to Core, can substantially improve multi-NPU LLM inference performance under identical hardware conditions.